Chapter 14: Incomplete Information, Static Games

ECON 5001
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Based on: *Game Theory* by Giacomo Bonanno

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Where We're Going

So far: the game is **common knowledge**

Now: some player isn't sure **which game** they're playing

- Usually: unsure about someone else's **payoffs**
- "Is my rival's cost high or low?"

Tools we have:

- Knowledge-belief structures (deck 11): who knows what
- Bayesian updating (deck 12): beliefs from strategies

New tool: the **Harsanyi transformation**

MobLab: “Market for Lemons”

- MobLab time!
- You’re a **seller** or a **buyer** of a used car
- Seller knows what the car is worth to them: between $500 and $5,000
- Seller names a price. Buyer accepts or rejects
- Buyer doesn’t see the car’s value until after
- The car is worth 50% more to the buyer than to the seller

Feel free to talk! And play it the way you’d actually play it.

Imperfect vs. Incomplete Information

Imperfect information (Chapter 4)

- You don't know what others **did** earlier
- Information sets with more than one node

Incomplete information (now)

- You don't know what **game** you're playing
- Someone's payoffs (or actions, or...) are uncertain

Ex: someone says you owe them money and threatens to sue

- Will they really sue? Depends on their **type**: aggressive or not
- You don't know their type. They do

One-Sided Incomplete Information

Example: Two Possible Games

P1 is type *a*

1 \ 2	L	R
T	0, 3	3, 9
B	3, 3	0, 0

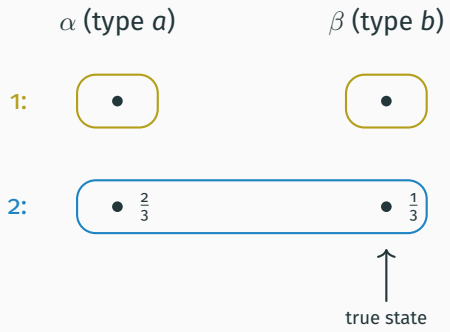
P1 is type *b* (the true game)

1 \ 2	L	R
T	6, 3	0, 9
B	3, 3	3, 0

- Only P1's payoffs differ
- P1 knows which game it is
- P2 thinks: type *a* with prob. $\frac{2}{3}$, type *b* with prob. $\frac{1}{3}$
- All of this is common knowledge

Bonanno Figures 14.1–14.2.

The Knowledge-Belief Structure



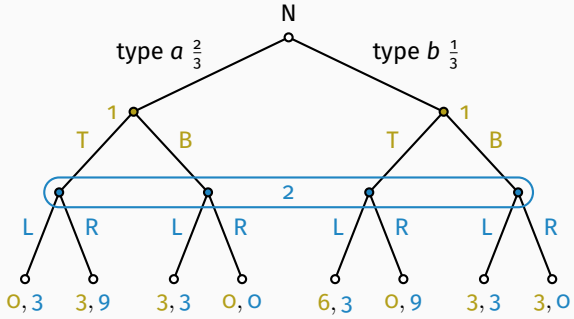
Each state comes with a **game**: $\alpha \to$ type a's game, $\beta \to$ type b's game

What's common knowledge? (Constant across states!)

- _____ _____ _____

Bonanno Figure 14.3 draws the two games swapped. This is the correct version.

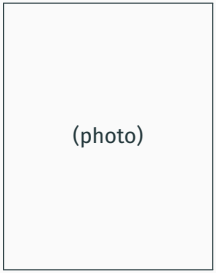
The Harsanyi Transformation



1. **Nature** picks the state
   - Prob. = P2's beliefs
2. P1 learns it, chooses T or B
3. P2 chooses L or R **without** seeing either

Now it's a regular game! Imperfect information, but **complete**

John Harsanyi (1920–2000)



- Budapest. Same high school as von Neumann
- 1944: escaped a deportation train at the station
- 1950: snuck across a marsh into Austria
- Sydney: factory job by day, economics at night
- Games of incomplete information: 1967–68, three papers
- 1994 Nobel Prize, with Nash and Selten

Not in the book. Harsanyi's Nobel autobiography (nobelprize.org).

Strategies and the Strategic Form

P1 has two information sets (one per type) $\Rightarrow$ strategy $=$ (choice if a, choice if b)

Player 1 \ Player 2	L	R
TT	$2, 3$	$2, 9$
TB	,	,
BT	,	,
BB	$3, 3$	$1, 0$

Payoffs are **expected** payoffs. **Ex:** TT vs. L, P1 gets $\frac{2}{3}(0) + \frac{1}{3}(6) = 2$

Fill in TB and BT

Strategies and the Strategic Form

P1 has two information sets (one per type) $\Rightarrow$ strategy $=$ (choice if a, choice if b)

Player 1 \ Player 2	L	R
TT	2, 3	2, 9
TB	1, 3	3, 6
BT	4, 3	0, 3
BB	3, 3	1, 0

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TB	$1, \underline{3}$	$\underline{3}, \underline{6}$
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Payoffs are **expected** payoffs. **Ex:** TT vs. L, P1 gets $\frac{2}{3}(0) + \frac{1}{3}(6) = 2$

Bayesian Nash Equilibrium

Definition
A **Bayesian Nash equilibrium** (BNE) is a NE of the game from the Harsanyi transformation

"Bayesian" just reminds us it came from incomplete information

Our example: pure BNE are _____ and _____

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Our example: pure BNE are _____ and _____

Fact: the Harsanyi game is finite $\Rightarrow$ a BNE always exists (maybe mixed)

What Does a BNE Mean?

Take the BNE $((T, B), R)$. The true state is β (type b)

- What actually gets played? _____
- Is that a NE of the true game? _____

Lesson: a BNE need not be a NE of the game actually being played

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But P1 **knows** she's type b. Why plan for type a??

- Her strategy $(T, B) =$ what she actually does (B)
  $+$ what P2 **thinks** type a would do (T)

Pooling vs. Separating

Definition (Bonanno 14.2.1)
In a pure BNE, the informed player is

- **Pooling** if every type makes the **same** choice
- **Separating** if different types make **different** choices

Our BNE: $((T, B), R)$ is _____ $((B, T), L)$ is _____

Pooling vs. Separating

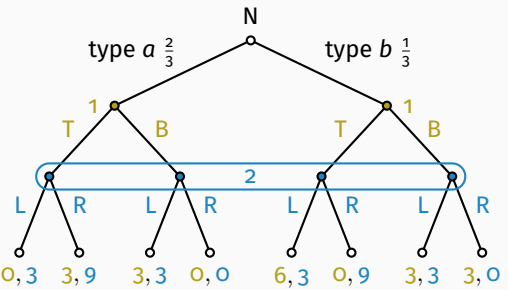
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Our BNE: $((T, B), R)$ is _____ $((B, T), L)$ is _____

Separating: P1's choice would **reveal** her type, if P2 could see it

Shortcut: Check Type by Type



Is $((B, T), L)$ a BNE?

P2's beliefs (Bayes):
(type a, B): _____ (type b, T): _____

P2: L gives _____ R gives _____

Type a vs. L: T ____ B ____

Type b vs. L: T ____ B ____

This is WSE logic: Bayesian updating + best replies. Bonanno uses it a lot

WSE $\Rightarrow$ NE (Theorem 11.3.1), so a WSE of the Harsanyi game gives a BNE.

Shell vs. Speedway, Again

Cournot (deck 04): Shell and Speedway choose gallons q_1, q_2. $P = 34 - Q$

- Shell's cost: \$6/gallon. Common knowledge
- Speedway's cost: really \$9/gallon (High)
- But Shell thinks: High ($\frac{1}{3}$) or a new low-cost tech, \$3 (Low, $\frac{2}{3}$)
- Speedway knows its own cost

Benchmark, complete info (both know $c_2 = 9$):

- Shell: $\max_{q_1} (34 - q_1 - q_2)q_1 - 6q_1$ Speedway: $\max_{q_2} (34 - q_1 - q_2)q_2 - 9q_2$
- $q_1 =$ _____ $q_2 =$ _____

Bonanno §14.2. Uses calculus, like deck 04.

Cournot BNE

Strategy for Speedway: one output **per type**: q_H and q_L

High Speedway: $\max_{q_H} (34 - q_1 - q_H)q_H - 9q_H \Rightarrow$ _____

Low Speedway: $\max_{q_L} (34 - q_1 - q_L)q_L - 3q_L \Rightarrow$ _____

Shell, who doesn't know the type:

$$\max_{q_1} \frac{1}{3}\left[(34 - q_1 - q_H)q_1 - 6q_1\right] + \frac{2}{3}\left[(34 - q_1 - q_L)q_1 - 6q_1\right]$$

$\Rightarrow$ _____

Cournot BNE

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$\Rightarrow$ _____

Solve: $q_1 =$ _____ $q_H =$ _____ $q_L =$ _____

Compare

	Complete info	Incomplete info
Shell	10.33	
Speedway (really High)	7.33	

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Shell fears a low-cost (aggressive) rival $\Rightarrow$ Shell produces _____

So High Speedway faces less competition $\Rightarrow$ produces _____

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Lesson: beliefs about your type change what others do, even when they're wrong

This Is Everywhere

Bayesian games: you know your own type, not theirs

- **Auctions:** you know your value for the item, not others'
- **Used cars:** the seller knows if it's a lemon
- **Insurance:** you know how risky you are. The insurer doesn't
- **Job interviews:** you know how good you are
- **Poker:** you know your cards
- **Negotiation:** you know your walk-away price

Real versions: many types, continuous types, many players, repeated play

- Same idea: each type best-responds to beliefs about others' types

Back to MobLab: Lemons

% of cars sold	
% of buyers who lost money	

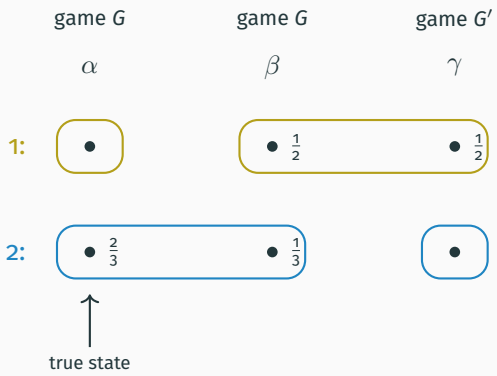
Akerlof's logic: say seller values are spread evenly from \$500 to \$5,000

- At price p: only cars worth $\leq p$ to their seller get offered
- Avg. value to the buyer: $1.5 \times \frac{500+p}{2}$. Worth paying p only if $p \leq$ _____
- Only _____% of cars trade, though **every** trade would help both!

Not in the book. Akerlof (1970, *QJE*). 2001 Nobel Prize.

Two-Sided Incomplete Information

Example: Both Players Unsure



Game G

1 \ 2	C	D
A	1, 3	0, 1
B	0, 0	3, 1

Game G'

1 \ 2	C	D
A	1, 0	0, 2
B	0, 0	3, 1

At α: both know it's G. But P2 isn't sure P1 knows!

Which Probabilities for Nature?

One-sided: use the uninformed player's beliefs. Two-sided: **two** sets of beliefs!

- P1 puts $\frac{1}{2}$ on β. P2 puts $\frac{1}{3}$ on β. Which one?

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Neither! Use the **common prior** (deck 11): one p that gives both players' beliefs

$p(\alpha) =$ _____ $p(\beta) =$ _____ $p(\gamma) =$ _____

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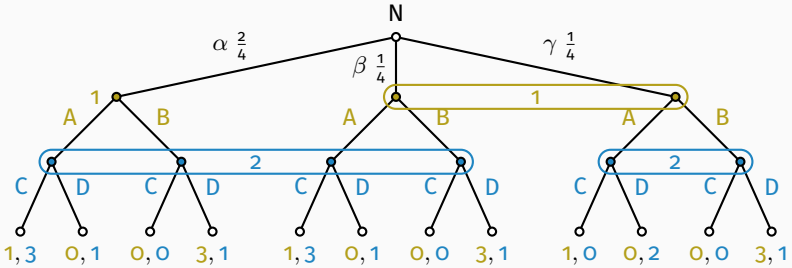
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$p(\alpha) =$ _____ $p(\beta) =$ _____ $p(\gamma) =$ _____

No common prior? Then no Harsanyi transformation (Bonanno Exercise 14.6)

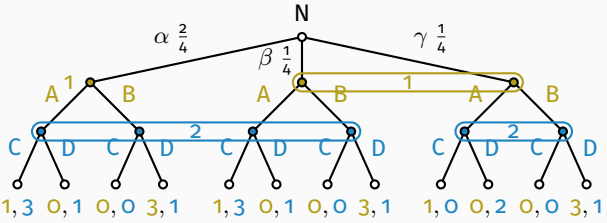
The Harsanyi Game



P1 knows if the state is α. P2 knows if the state is γ

Bonanno Figure 14.9.

Checking a BNE



Candidate:

P1: A at α, B at $\{\beta, \gamma\}$

P2: C at $\{\alpha, \beta\}$, D at $\{\gamma\}$

P2's left info set. Beliefs: _____

P2: C gives ____, D gives ____

P1 at $\{\beta, \gamma\}$: A ____, B ____

Three or more players (§14.4): same steps, more bookkeeping.

Let’s Work One Out

Who Brings the Grill?

Jordan and Casey tailgate before the game. Each chooses: **Bring** a grill or **Not**

- If at least one grill shows up: each gets 4. Otherwise 0
- Bringing costs 1 if you own a grill (**Low** type), 5 if you'd rent one (**High** type)
- You know your own type. Each is Low with prob. $\frac{1}{2}$, independently
- Actually both are Low

Ex: Jordan Low brings, Casey doesn't: Jordan gets $4 - 1 = 3$, Casey gets 4

Choices are simultaneous.

Who Brings the Grill?

1. Draw the knowledge-belief structure (4 states). What's the common prior?

2. How many pure strategies does each player have in the Harsanyi game?

3. Show that a High type never brings a grill

4. Find all pure BNE. Pooling or separating?

5. What happens at the true state? Is it a NE of the true game?

6. Now Pr(Low) $= 0.9$. Is your BNE still a BNE? If not, find one

What Have We Learned, Charlie Brown?

- **Incomplete info**: unsure which game. **Imperfect**: unsure what others did
- Model it with a knowledge-belief structure: one game per state
- **Harsanyi transformation**: Nature picks the state using the common prior
- **BNE** $=$ NE of the Harsanyi game
  - Strategy: one choice per type. Check type by type
  - Pooling vs. separating
- A BNE may not be a NE of the true game
- Two-sided: needs a common prior

Reading & Practice

- Bonanno, Chapter 14
  - §14.1–14.3: read. §14.4: skim
  - Figure 14.3 has the two games swapped (see errata)
- Exercises: §14.5.1 (14.1, 14.2), §14.5.2 (14.3–14.5), §14.5.3 (14.6)
- Skip 14.7 (three players) and 14.8 (challenging)
- Solutions in §14.6. Try them first